Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

The twelve monthly 2023 yellow taxi parquet files were sampled using frac = 0.007 and combined into one dataframe. This gave a total of 265,487 records.

Sampling was done from each file first, and then the sample parts were concatenated, which helps to run the operation smoothly, but covers all the months of 2023.

The main reason for sampling is to reduce time and memory usage. For a count-based result, the sample values can be scaled using the sampling fraction to get an approximate estimate of the full data.

Conclusion: The final sample is small enough to run smoothly, but still covers all months of 2023.

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

The index was reset to make a continuous index after combining 12 monthly files. I also dropped the store_and_fwd_flag column because it was not required for the main analysis. This column represents whether the trip record was stored before sending it to the vendor or not. But it was not useful to analyse demand, pricing, routing, zones, or customer experience.

2.1.2. Combine the two airport_fee columns

Both airport_fee and Airport_fee were present due to a naming inconsistency. I checked that both columns were not filled together, then combined them into a single airport_fee column and removed the duplicate column.

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

Missing value percentages were calculated for all columns. Airport_fee, congestion_surcharge, passenger_count, and RatecodeID had around 3.33% missing values, while the other important trip and fare columns did not show missing values.

2.2.2. Handling missing values in passenger_count

The passenger_count column has fixed, discrete numeric values so missing values can be imputed using mode.

Since a valid taxi trip should have at least 1 passenger. So, the passenger_count values equal to 0 were first replaced with NaN and then imputed with the mode value.

2.2.3. Handle missing values in RatecodeID

The RatecodeID column also has discrete numerical values with a fixed range (1 to 6) considered as a categorical column. So, missing values were imputed with the mode as the most common rate code value in the column.

2.2.4. Impute NaN in congestion_surcharge

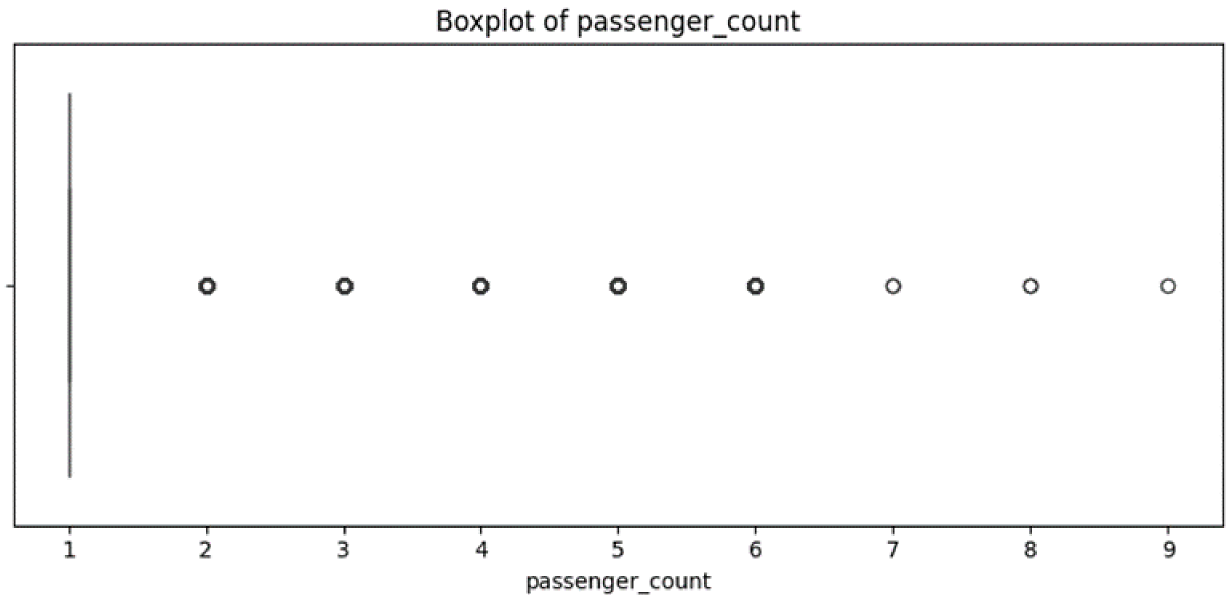
Since congestion_surcharge has only a few fixed values and 2.5 is the most frequent value, missing values were imputed using the mode.

2.3. Handling Outliers and Standardising Values

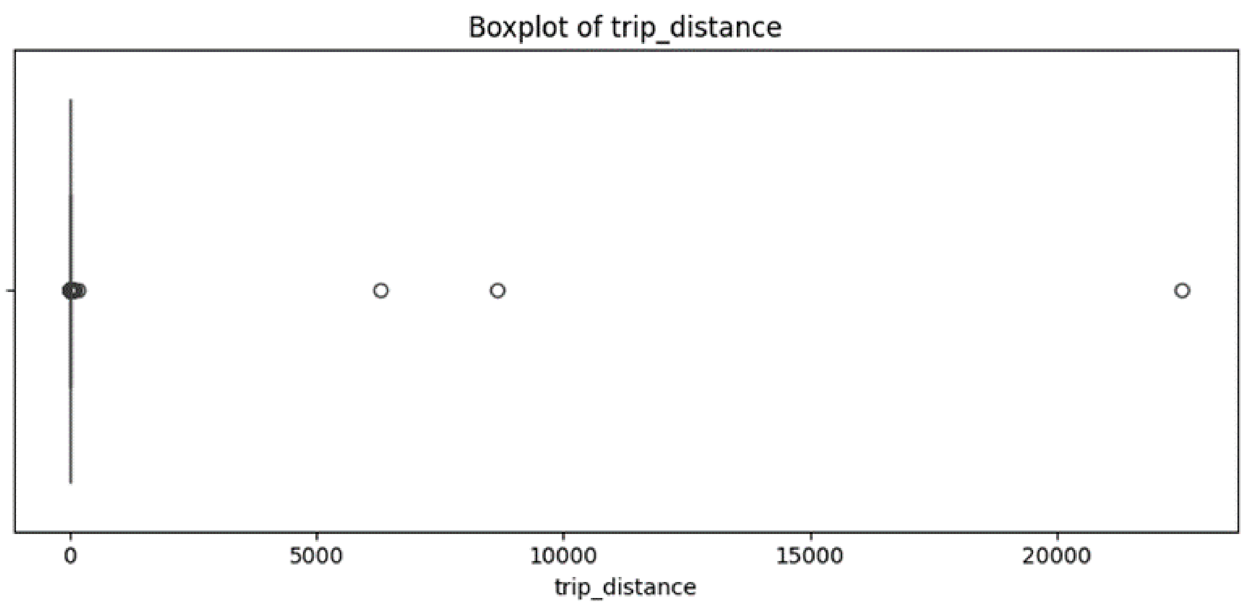
2.3.1. Check outliers in payment type, trip distance and tip amount columns

Outliers are checked using various ways, min-max range, quantiles, BoxPlot, and IQR. Boxplots helped visually identify extreme values in passenger_count, trip_distance, fare_amount, tip_amount, tolls_amount and total_amount.

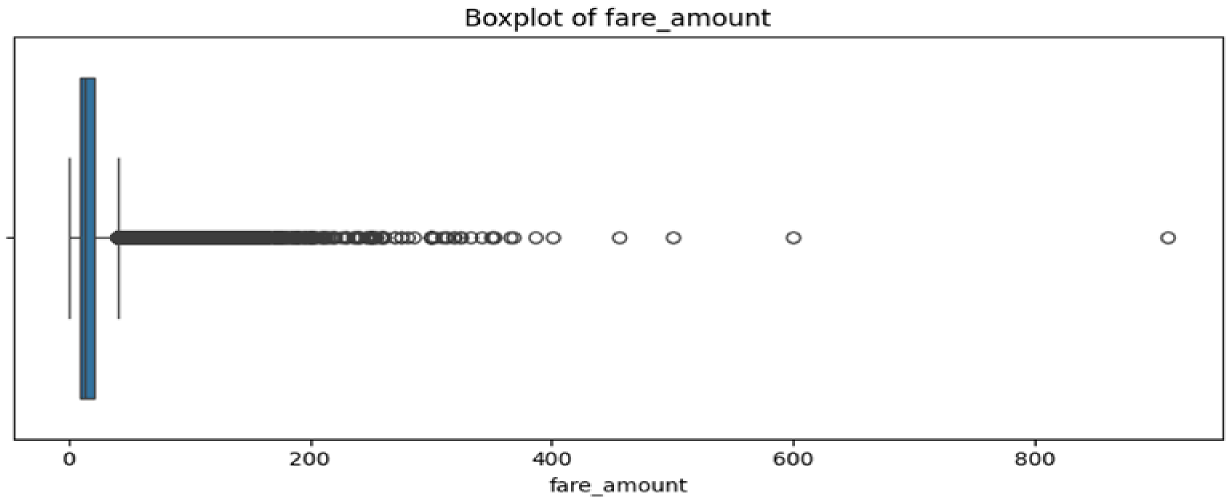
Passenger Count



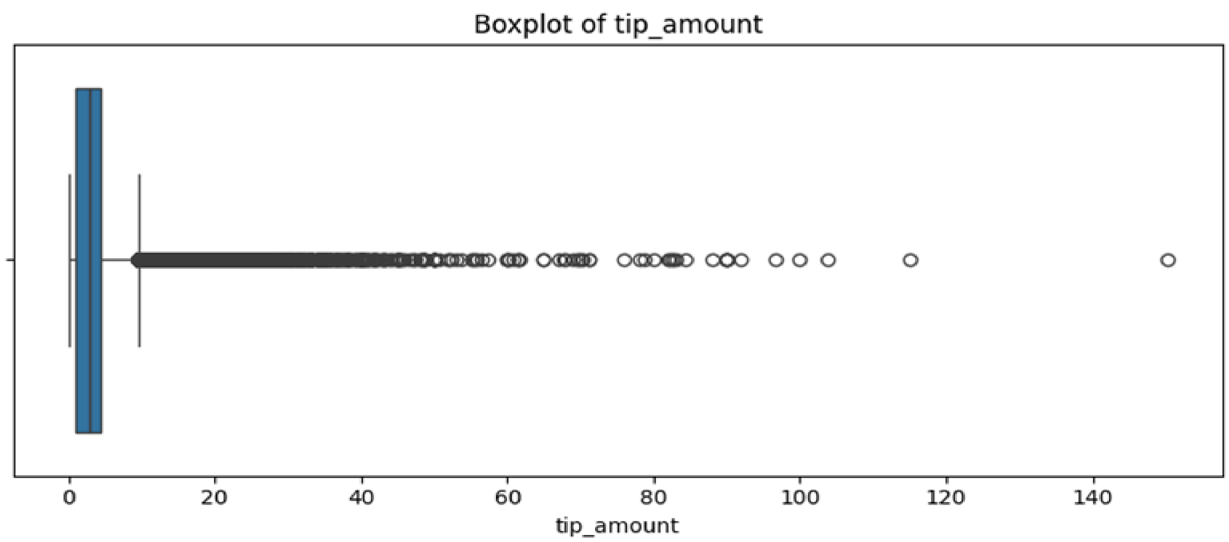
Trip Distance



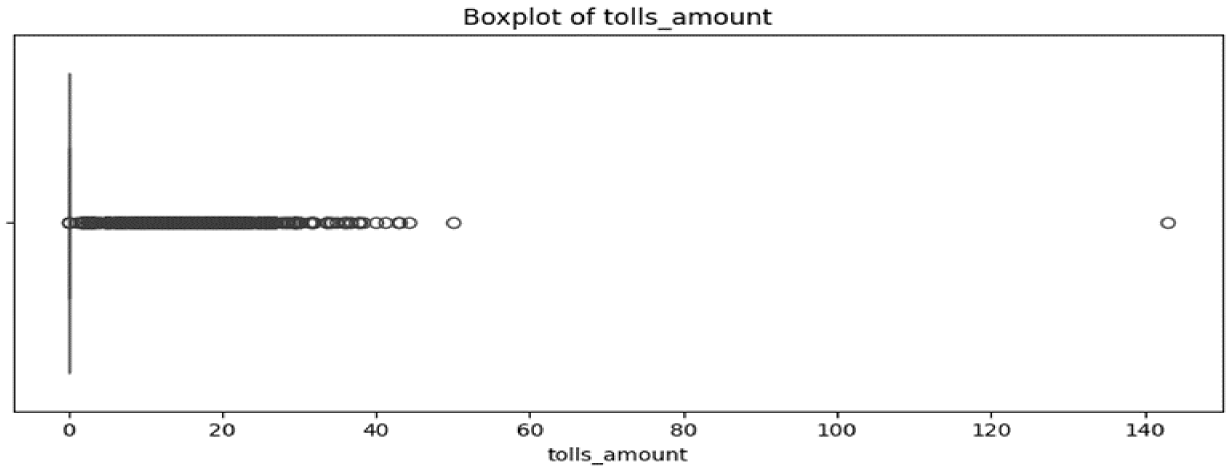
Fare Amount



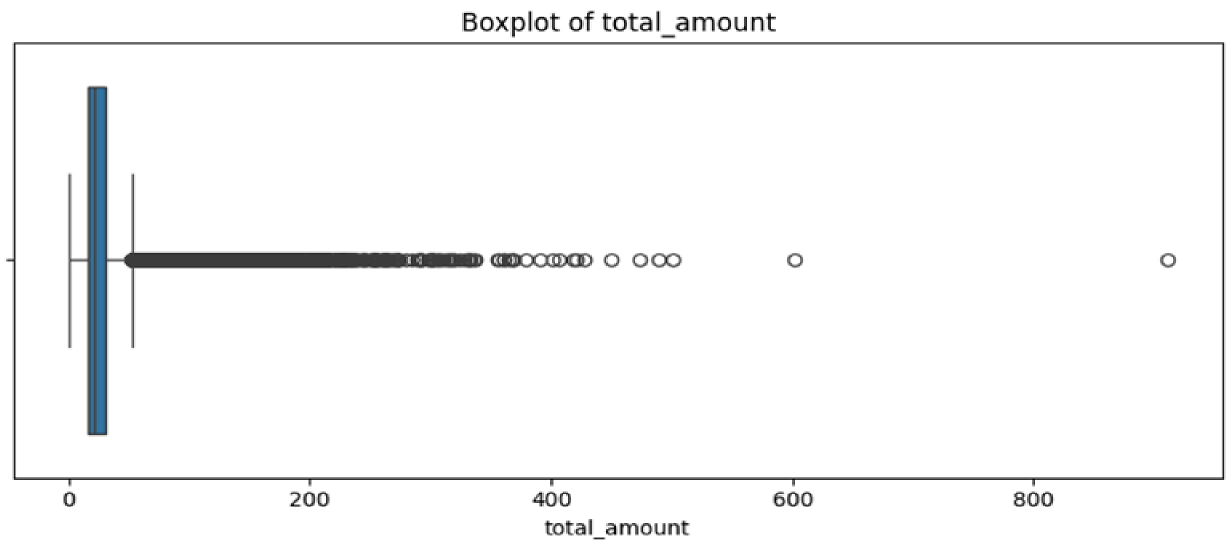
Tip Amount



Tolls Amount



Total Amount



IQR and Box Plots methods were used to identify outliers. However, not all outliers were directly removed because some high values can still be valid taxi trips.

One clearly invalid or unrealistic value was removed. For Example, in our NYC taxi dataset analysis, valid ranges such as payment_type should be between 1 and 6, RatecodeID should be between 1 and 6, and values outside of this considered invalid and outliers.

Conclusion: IQR and boxplots were used to identify possible outliers, but only clearly invalid or unrealistic records were removed. This kept valid long-distance, toll-related, and high-value trips in the dataset.

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

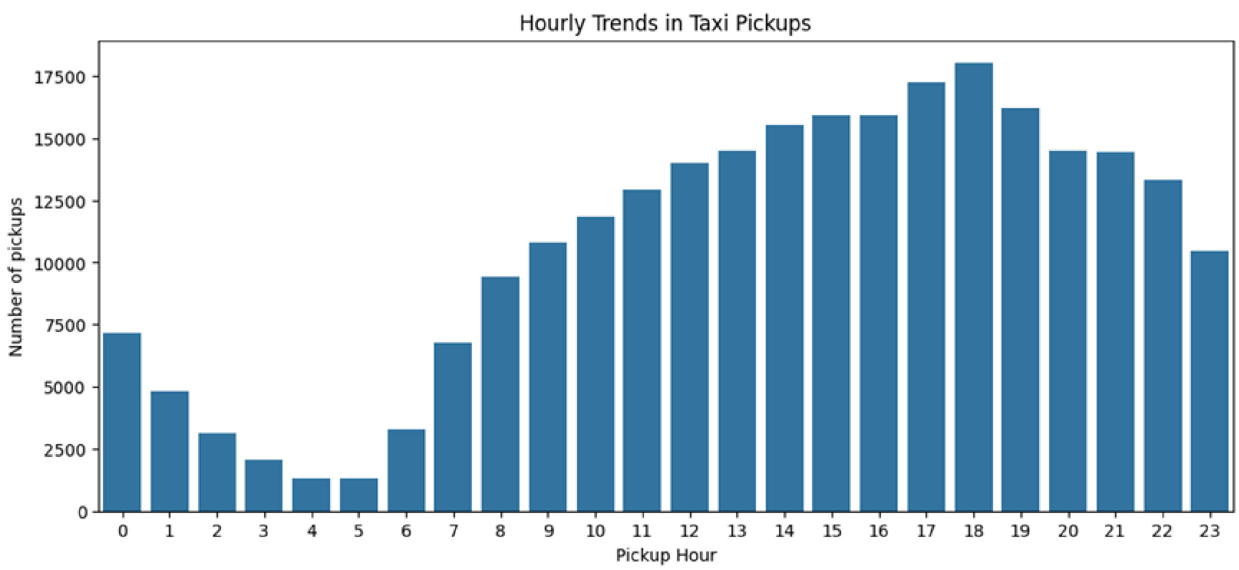
3.1.1. Classify variables into categorical and numerical

The columns were classified based on their role in the analysis. VendorID, RatecodeID, payment_type, and pickup/dropoff were treated as categorical variables, while distance, fare tip, total amount, trip duration, and surcharges values were treated as numerical variables.

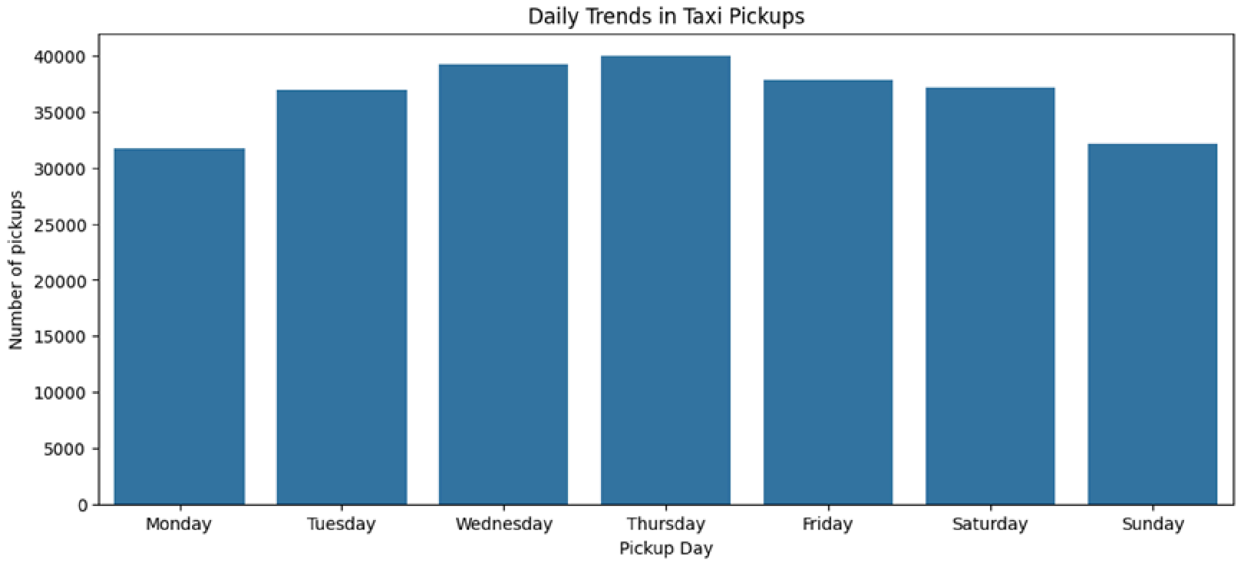
3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

The taxi pickup hour, day of the week, and months data were derived from the tpep_pickup_datetime column. This helped identify when taxi demand is higher and how pickup demand changes across different days and months of the year.

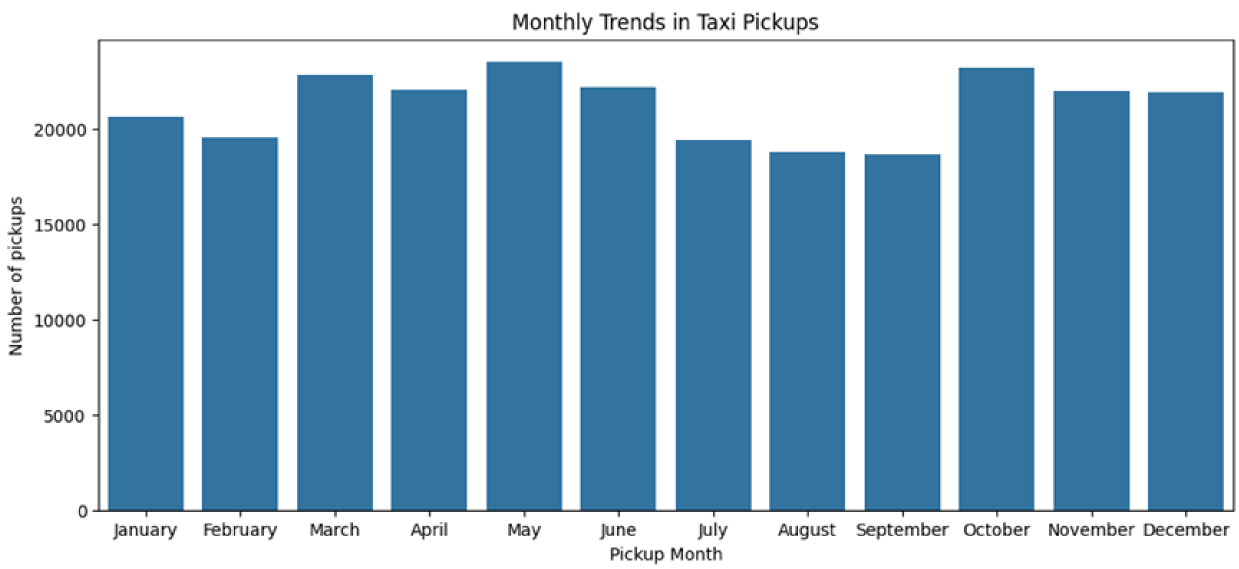
Taxi pickup distribution by hour



Taxi pickup distribution by week



Taci pickup distribution by month

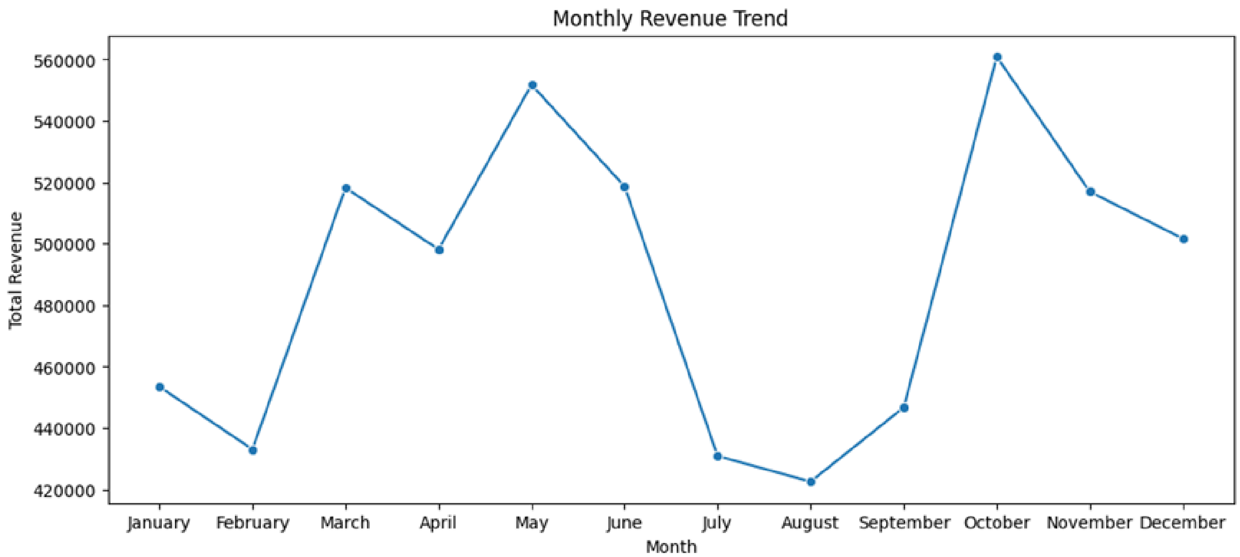


3.1.3. Filter out the zero/negative values in fares, distance and tips

For fare_amount, trip_distance, and tip-related analysis, invalid records were filtered using task-specific copies of the dataset. fare_amount and trip_distance were filtered to keep only positive valid values, while the tip_amount was allowed to be 0 because not every passenger gives a tip.

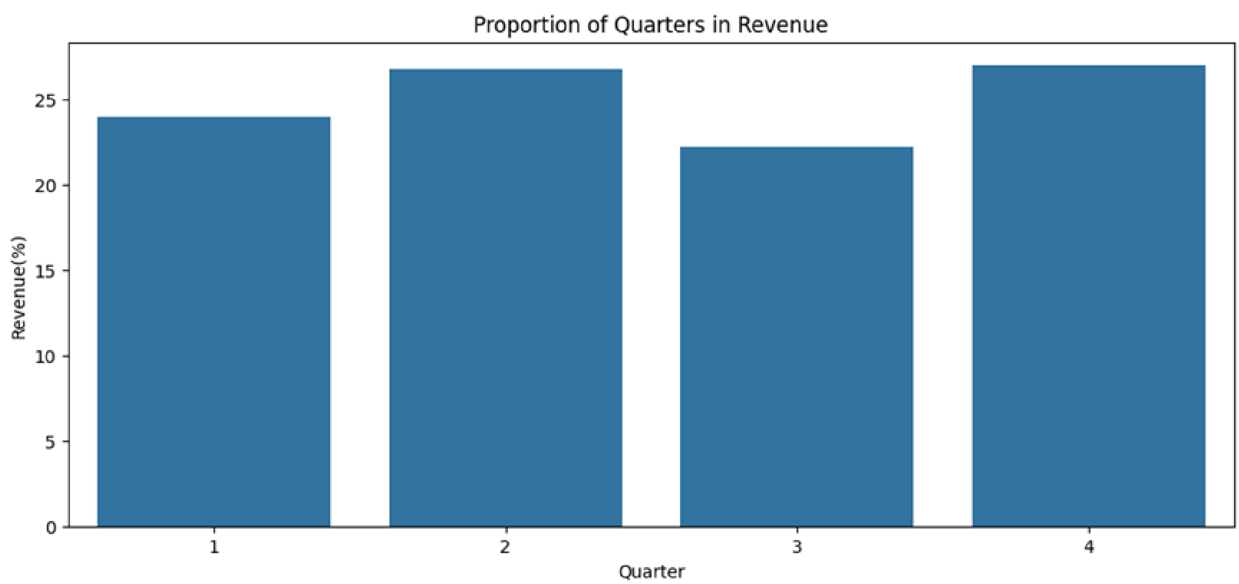
3.1.4. Analyse the monthly revenue trends

The monthly revenue was calculated by grouping total_amount by pickup_month. The line chart shows how revenue changes during the year and indicates that revenue is not evenly distributed across all months.



3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

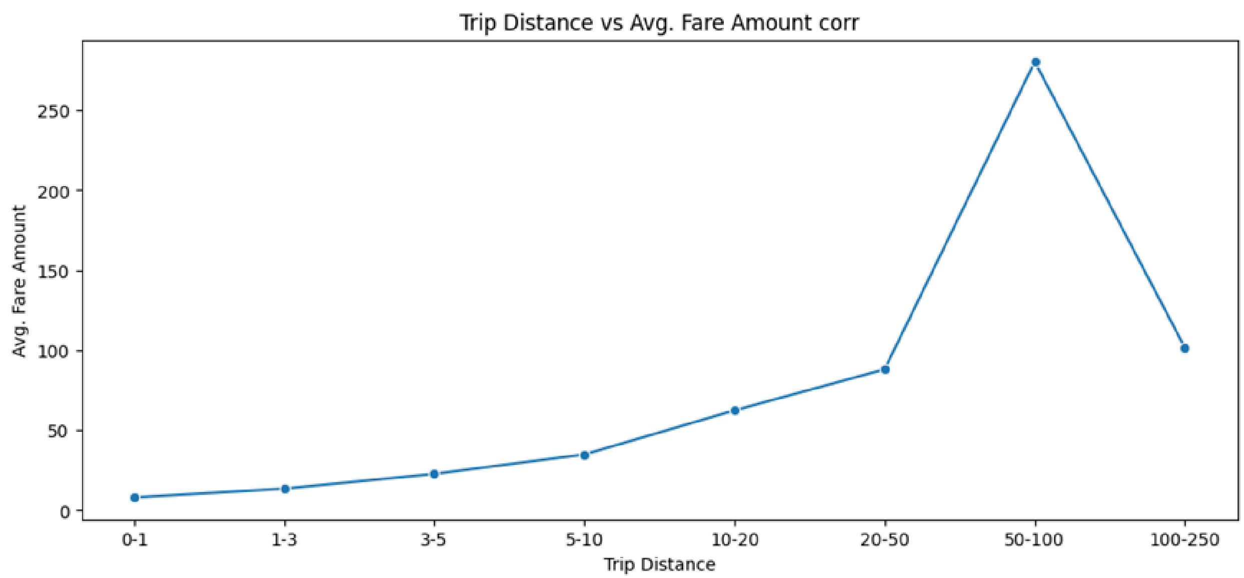
The revenue was grouped by quarter and divided by total yearly revenue. The Q2 and Q4 contributed the highest shares, while Q3 was comparatively lower.



3.1.6. Analyse and visualise the relationship between distance and fare amount

I have analysed the relationship between distance and fare amount using valid positive records. The taxi fare depends mainly on the time and distance. The chart clearly indicates that as the distance increases, the fare amount also increases.

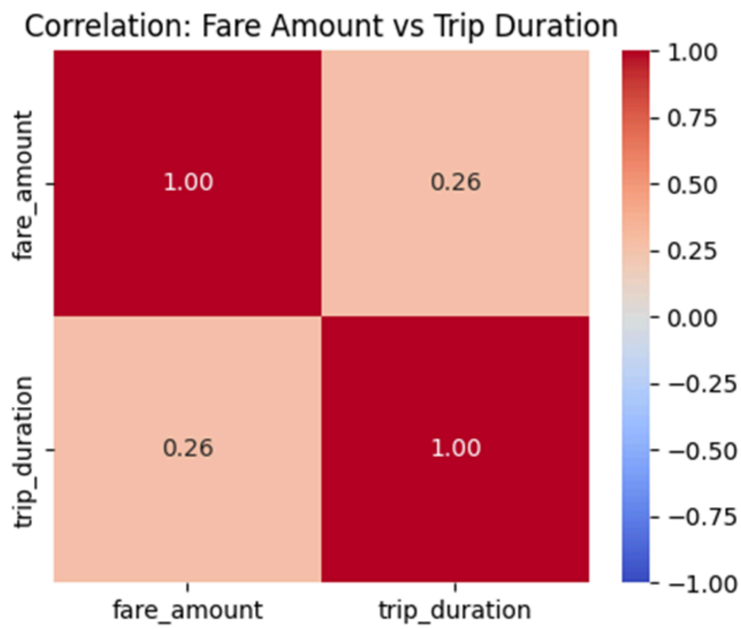
Average fare amount by trip distance range

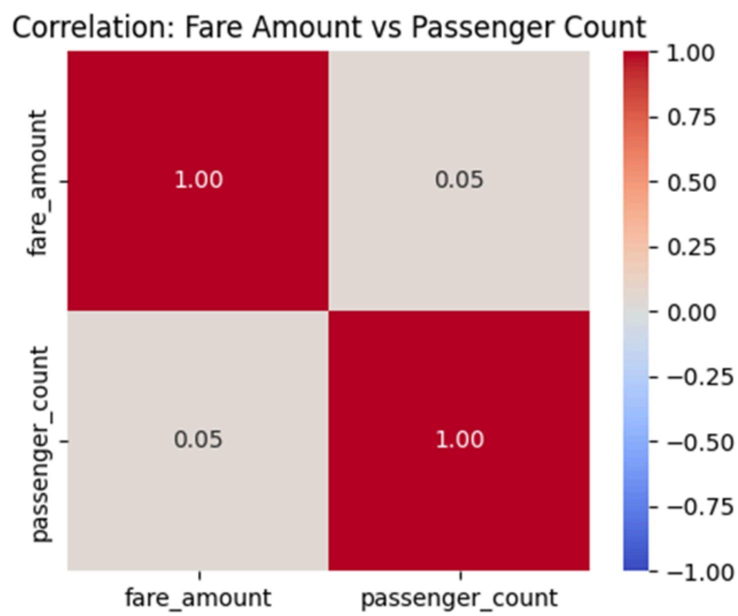


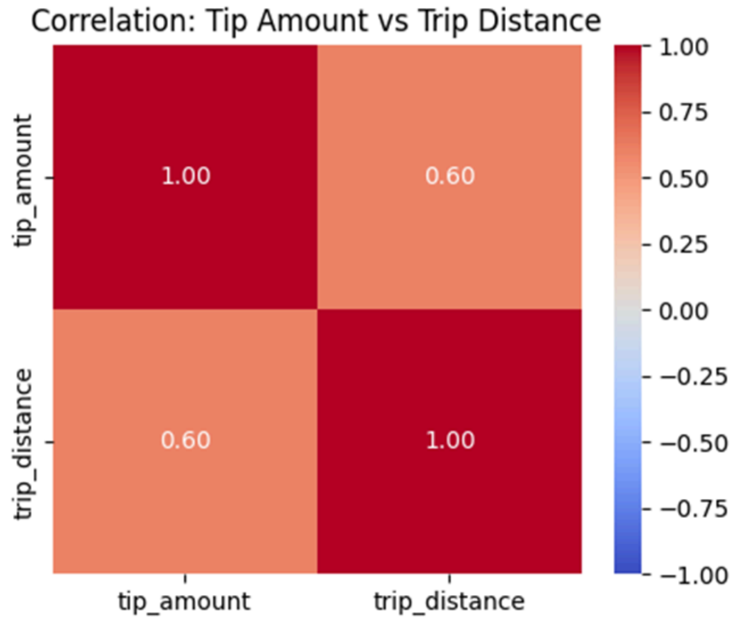
3.1.7. Analyse the relationship between fare/tips and trips/passengers

The pairwise correlations were calculated and visualised using heatmaps. Fare_amount showed a positive relation with trip_duration, which means longer trips usually have higher fares.

Fare_amount had a weak relation with passenger_count, so passenger count does not strongly affect fare. Tip_amount had a positive relation with trip_distance, but the relation was not very strong. So distance is only one of the factors affecting tips.

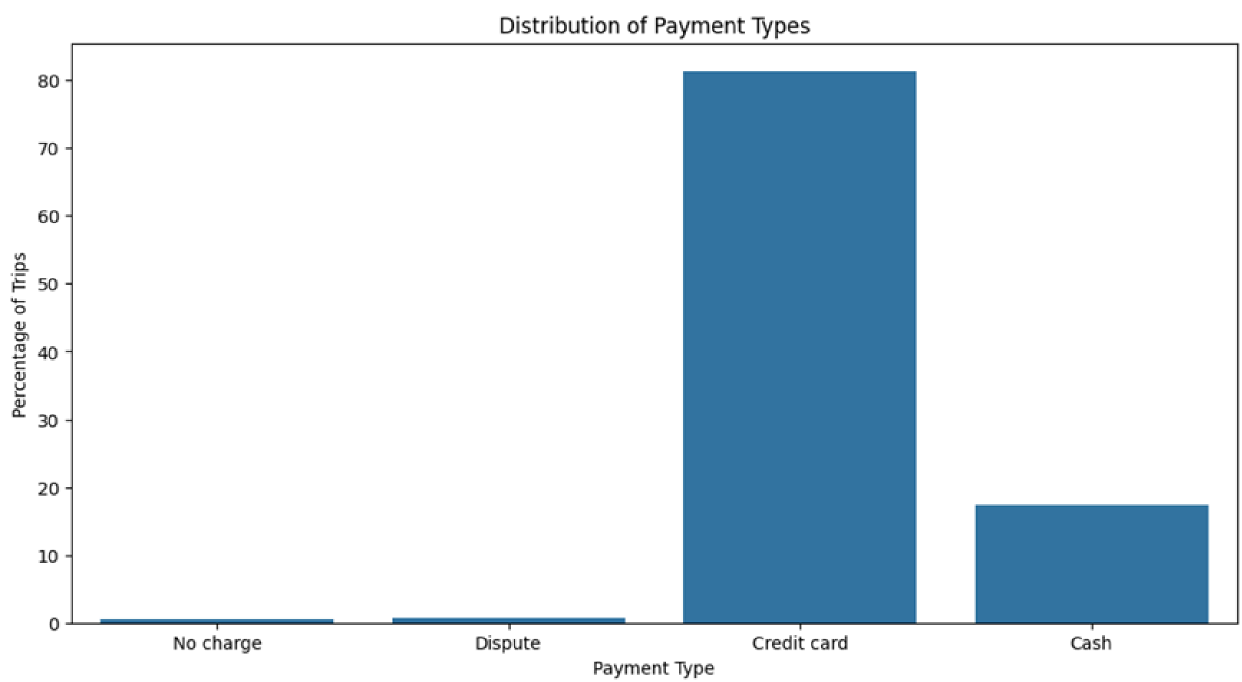

Correlation: Fare Amount vs Trip Duration
fare_amount
trip_duration
1.00
0.26
0.26
1.00
fare_amount
trip_duration
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Correlation: Fare Amount vs Passenger Count
fare_amount
passenger_count
1.00
0.05
0.05
1.00
fare_amount
passenger_count
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00



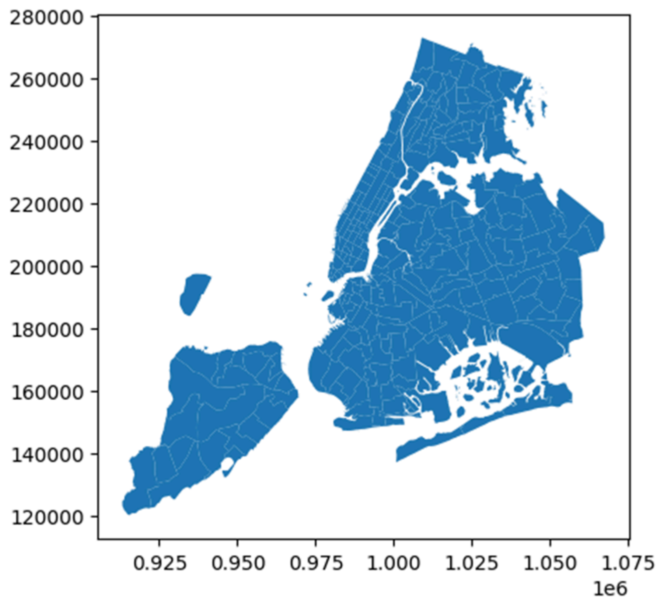
3.1.8. Analyse the distribution of different payment types

After removing invalid payment codes, payment type distribution was analysed, and Credit Card was the major payment mode, followed by cash, while other records had very small proportions.



3.1.9. Load the taxi zones shapefile and display it

For the geographical taxi zones analysis, the taxi zones shapefile was loaded. This shapefile helped connect trip records with pickup and dropoff zone names and boroughs.



3.1.10. Merge the zone data with trips data

Trip records were merged with zone data using PULocationID and DOLocationID. This created readable pickup and drop-off zone names for later zone-level analysis.

3.1.11. Find the number of trips for each zone/location ID

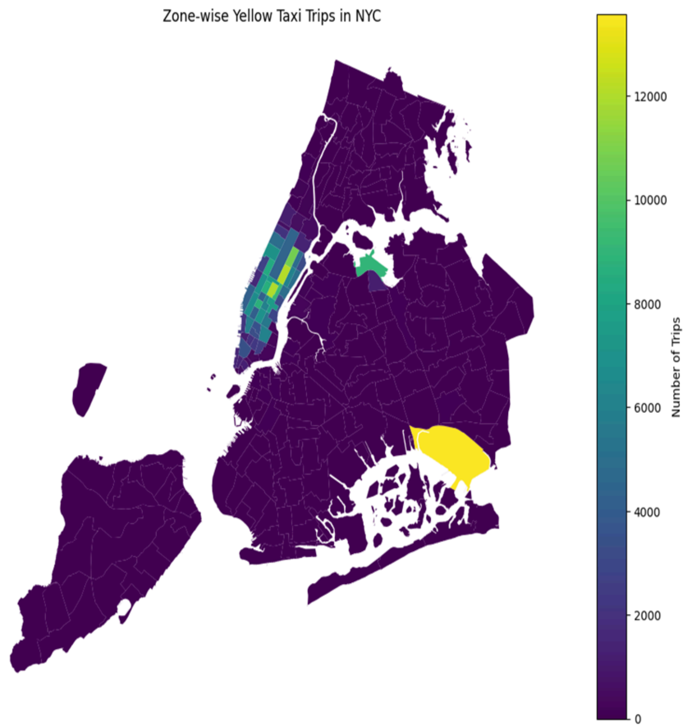
Trips were grouped by pickup LocationID to calculate trip volume for each zone. This helped find high-demand pickup locations based on the number of trips starting from each zone.

3.1.12. Add the number of trips for each zone to the zones dataframe

The trip_count values were merged into the zones dataframe. Missing trip counts were imputed with the value 0, so all zones could be displayed properly on the map.

3.1.13. Plot a map of the zones showing number of trips

The choropleth zone map was plotted using the number of trips in each zone. The map shows that trip demand is focused in selected zones rather than evenly spread across all NYC taxi zones.



3.1.14. Conclude with results

General EDA showed a clear variation in taxi pickup demand across hours, weekdays, and months. Revenue also varied by month and quarter, and trip activity was focused around selected high-demand zones.

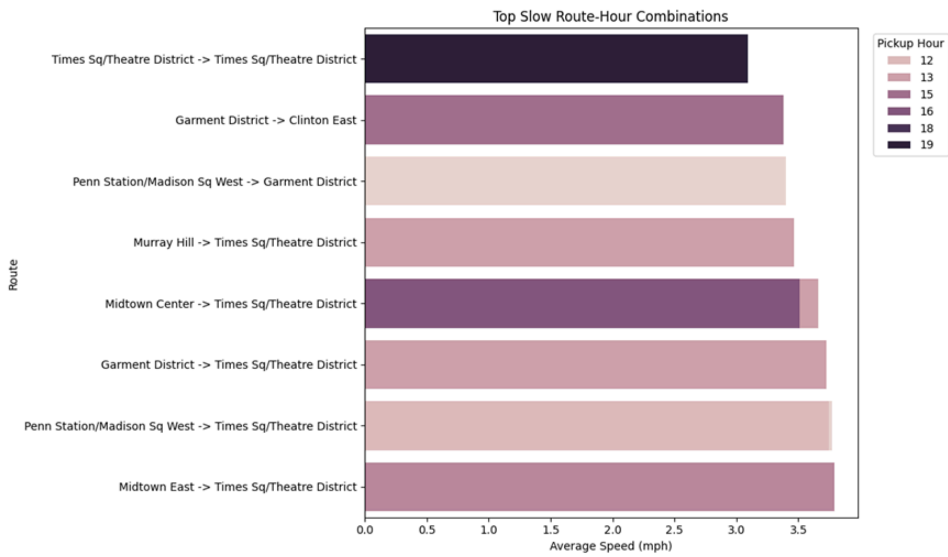
Trip fare and distance showed a positive correlation, while passenger_count didn't strongly relate to fare amount. These patterns were used for the detailed operational and pricing analysis.

Conclusion: The general EDA showed that pickup time, trip distance revenue period, payment type, and pickup/dropoff zones are useful variables for understanding taxi demand, revenue and trip behaviour.

3.2. Detailed EDA: Insights and Strategies

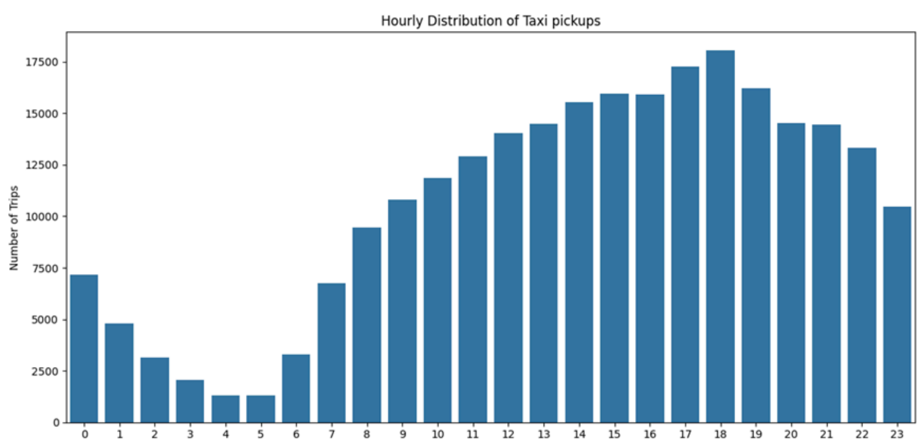
3.2.1. Identify slow routes by comparing average speeds on different routes

The slow route pairs were identified by calculating trip_duration and average speed for each route. Routes with low average speed were treated as inefficient because drivers may complete fewer trips compared to faster routes.



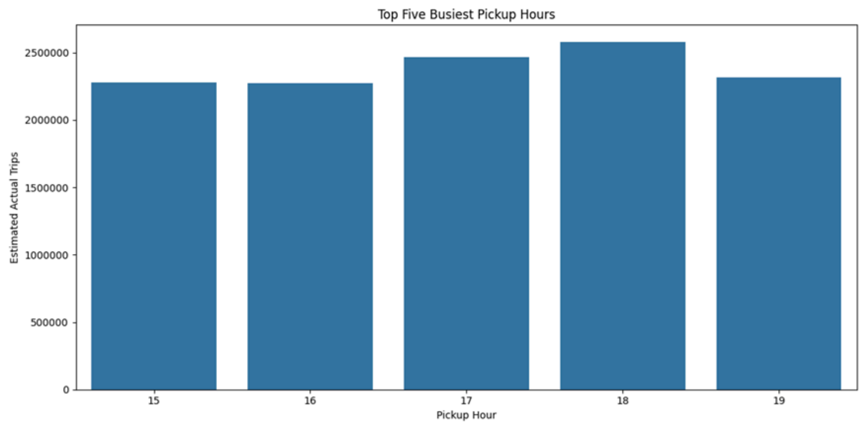
3.2.2. Calculate the hourly number of trips and identify the busy hours.

Hourly trip counts were calculated to identify busy periods. The busiest pickup hour was 18:00.



3.2.3. Scale up the number of trips from above to find the actual number of trips

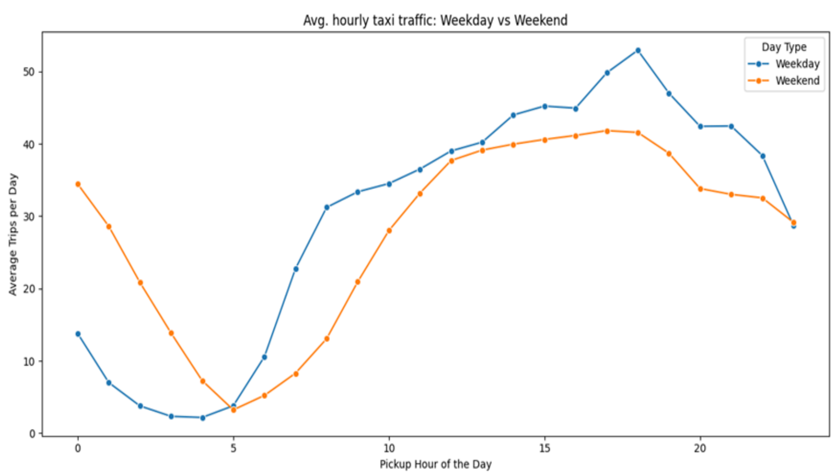
Since the sample data frac = 0.007, hourly trip counts were scaled by 1 / 0.007 to estimate actual trip counts. The highest estimated actual trips were around. 18:00.



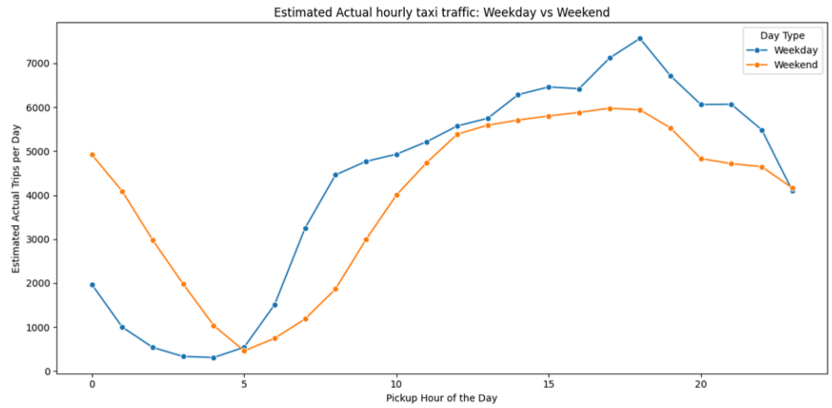
3.2.4. Compare hourly traffic on weekdays and weekends

As per the traffic hourly pattern analysis on weekdays and weekends separately. It helps to understand that dispatch planning shouldn't be on the same day, because weekend and weekday have different traffic patterns.

Hourly Traffic: Weekday vs Weekend



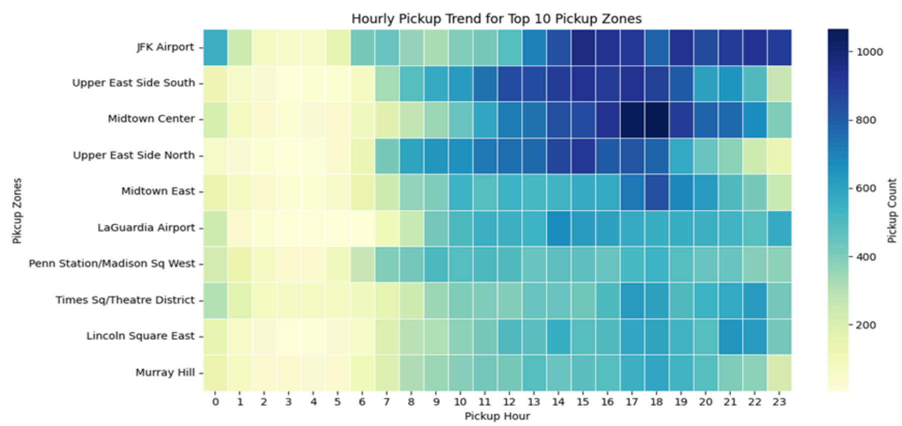
Estimated Actual Hourly Traffic: Weekday vs Weekend



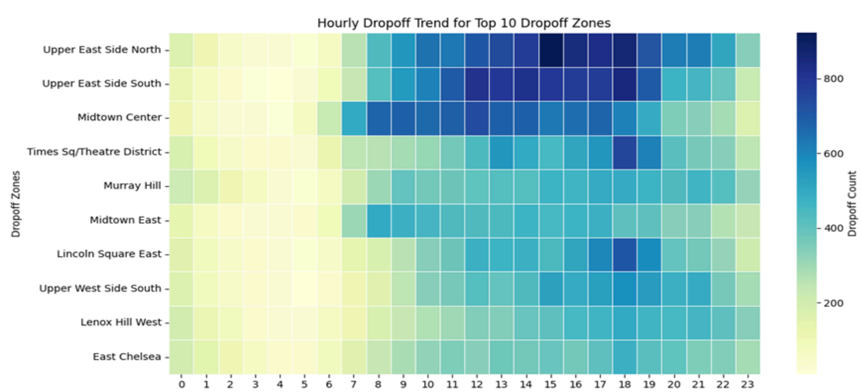
3.2.5. Identify the top 10 zones with high hourly pickups and drops

After analyzing the pickup and dropoff zones, I checked across hours. So, 'JFK Airport' was the highest pickup zone, and 'Upper East Side North' was the highest dropoff zone.

Hourly Analysis: Top 10 Pickup Zones



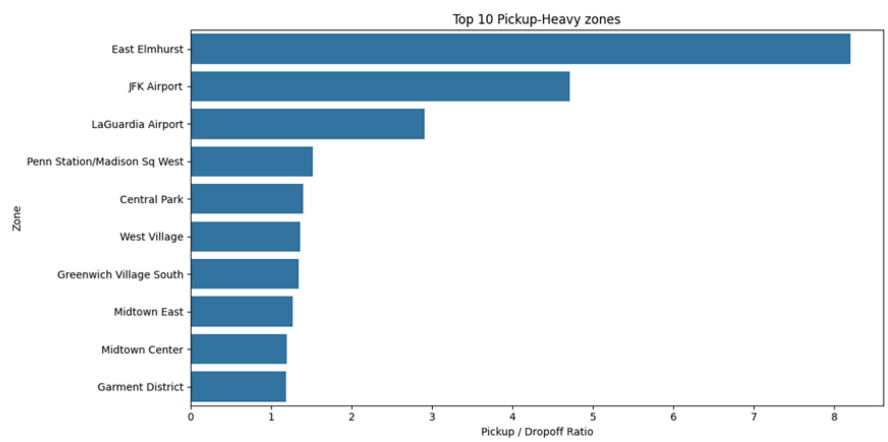
Hourly Analysis: Top 10 Dropoff Zones



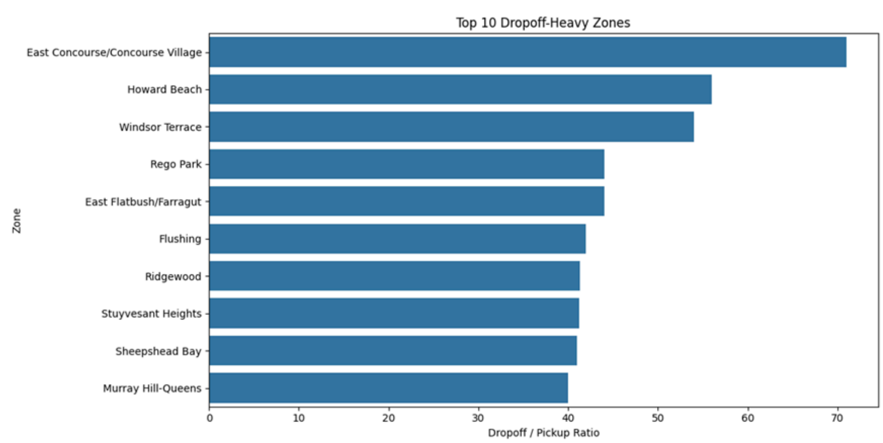
3.2.6. Find the ratio of pickups and dropoffs in each zone

After filtering records with positive pickup and dropoff counts calculated the ratio of the Pickup/dropoff for each zone. So, ‘East Elmhurst’ was the most pickup-heavy zone, while ‘East Concourse/Concourse village’ was the most dropoff-heavy zone based on the inverse ratio.

Top pickup-heavy zones by pickup/dropoff ratio



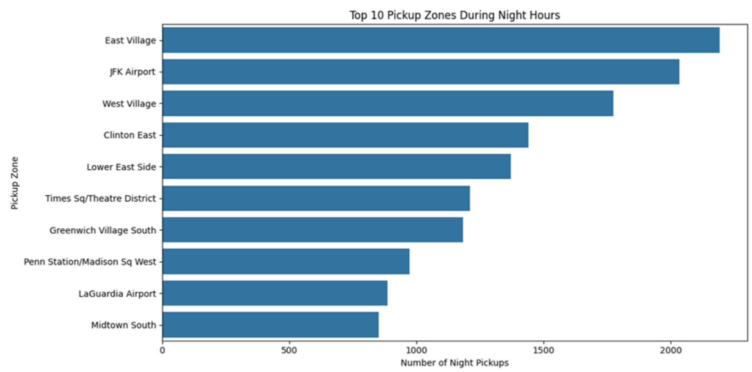
Top dropoff-heavy zones by pickup/dropoff ratio



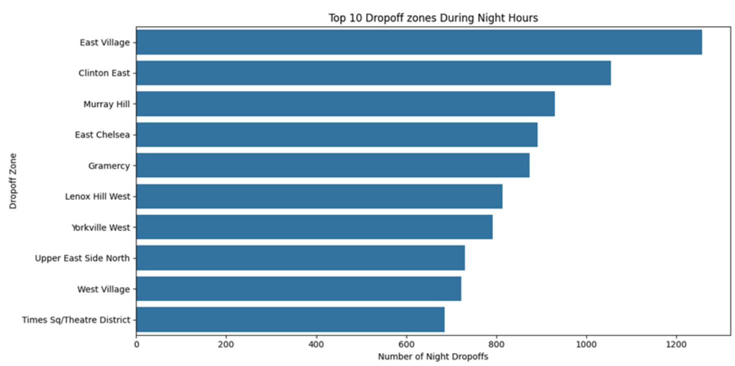
3.2.7. Identify the top zones with high traffic during night hours

Night Hours were defined between 11 PM and 5 AM. 'East Village' had the highest night pickup and dropoff traffic, so it is an important late-night demand zone.

Top 10 pickup zones



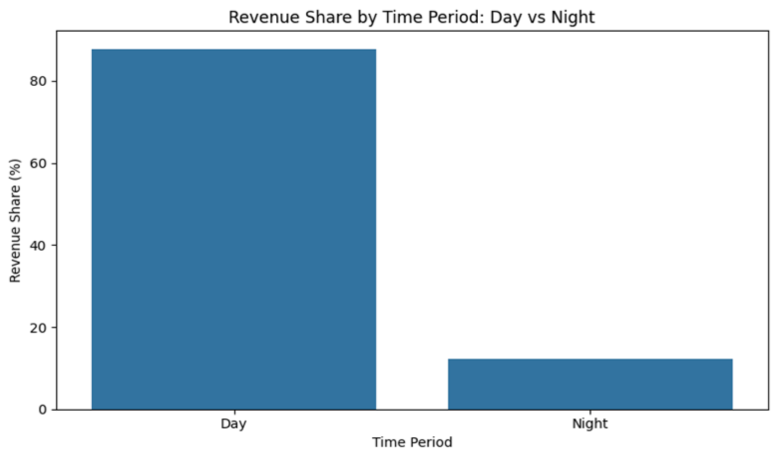
Top 10 pickup zones



3.2.8. Find the revenue share for nighttime and daytime hours

Daytime and nighttime revenue shares were calculated using total_amount. Daytime trips contributed 87.83% of the revenue, while nighttime trips contributed 12.17%.

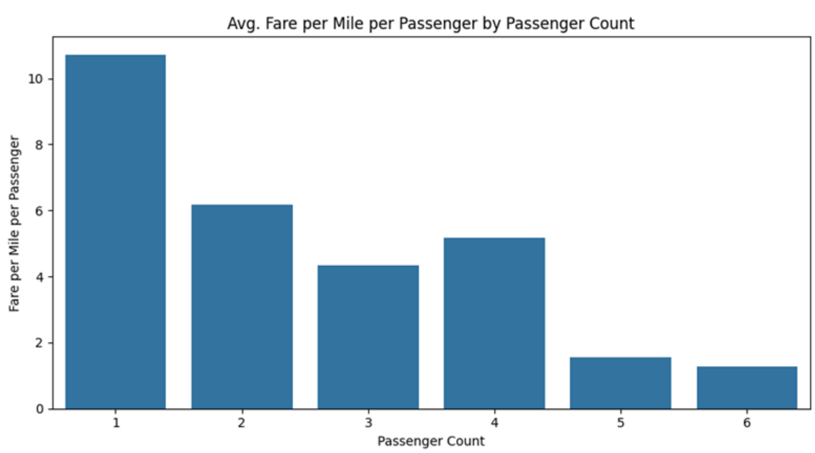
Revenue share by daytime and nighttime



3.2.9. For the different passenger counts, find the average fare per mile per passenger

Fare per mile was calculated first and then divided by passenger count 1 had the highest fare per mile per passenger, while the larger passenger group had a lower per-passenger cost.

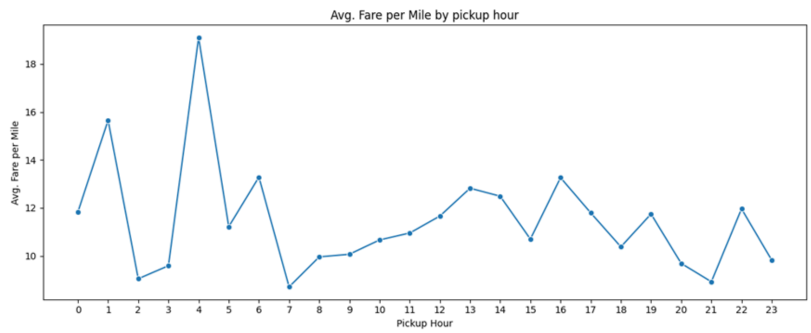
Fare per mile per passenger by passenger count



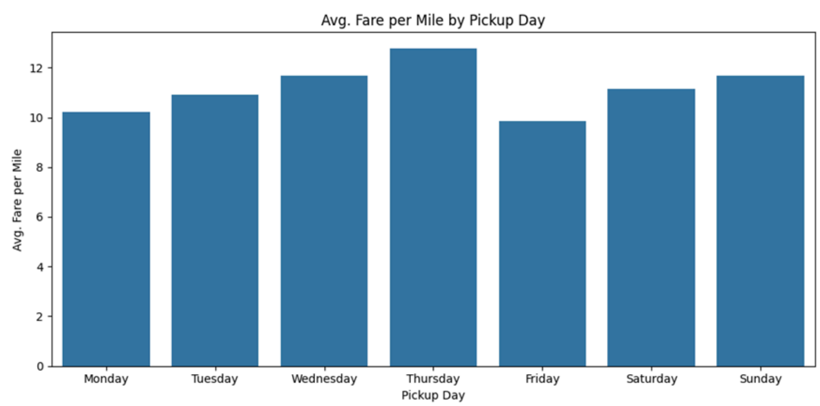
3.2.10. Find the average fare per mile by hours of the day and by days of the week

Average fare per mile was compared by pickup hour and weekday. The highest average fare per mile appeared around 4:00, and Thursday had the highest average fare by day.

Average fare per mile by pickup hour



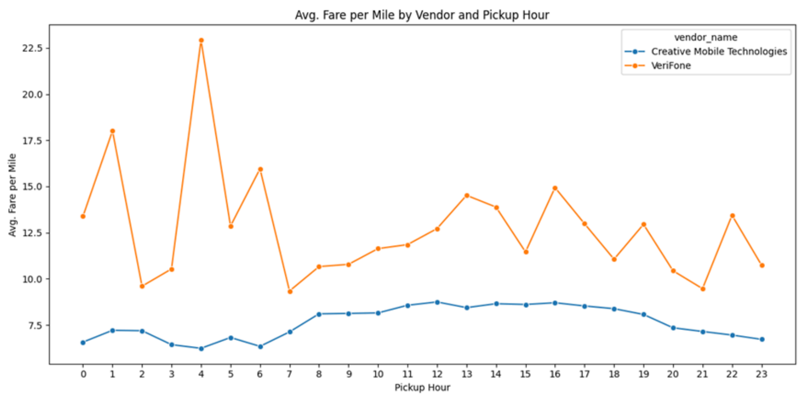
Average fare mile by pickup day



3.2.11. Analyse the average fare per mile for the different vendors

Vendor-wise fare per mile was analysed across pickup hours. 'VeriFone' had the highest fare per mile at 4:00. Show that vendor and time together can affect fare pattern.

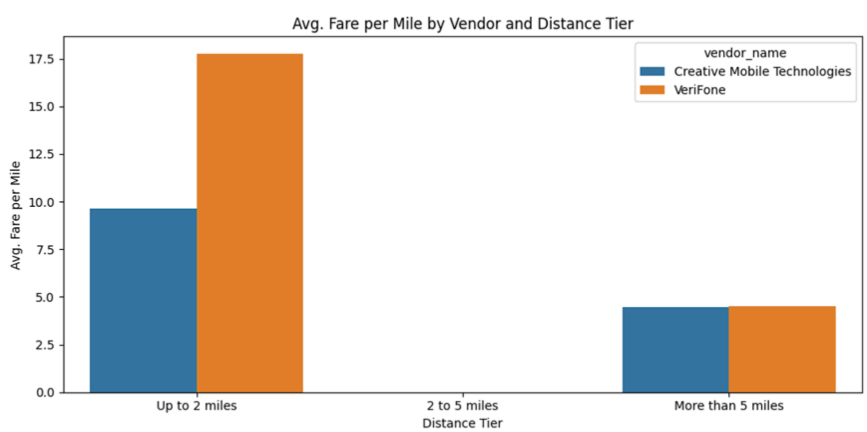
Average fare per mile by vendor and pickup hour



3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

Trips are divided into distance tier: up to 2 miles, 2 to 5 miles, and more than 5 miles. VeriFone had the highest average fare miles in the short-tier tier.

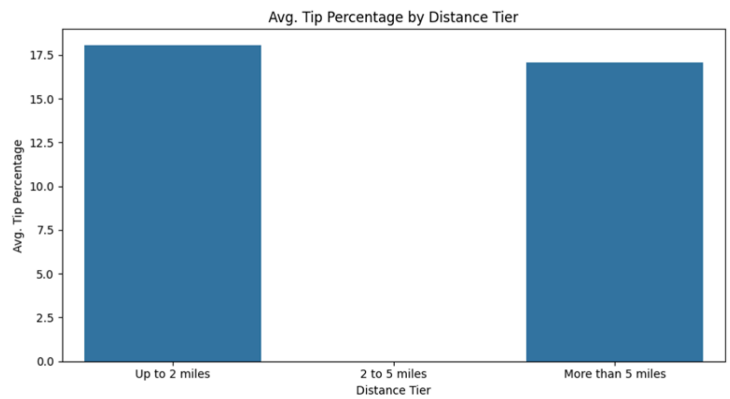
Average fare per mile by vendor and distance tier.



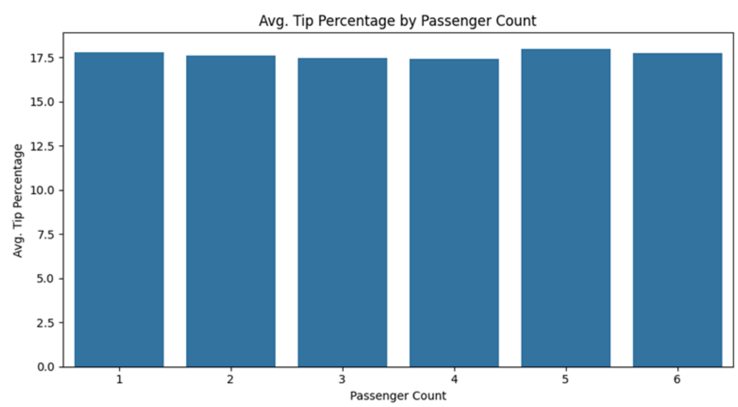
3.2.13. **Analyse the tip percentages**

The tip percentage was calculated using the credit-card trips because the cash tips are not captured in the dataset. Tip patterns were analysed by distance tier, passenger count, and pickup hour.

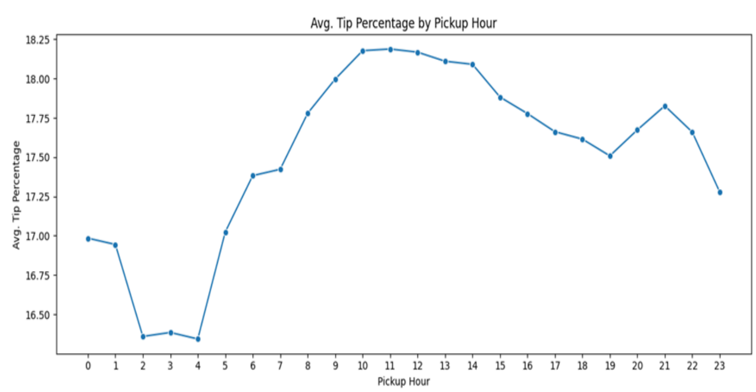
Average tip percentage by distance tier



Average tip percentage by passenger count



Average tip percentage by pickup hour

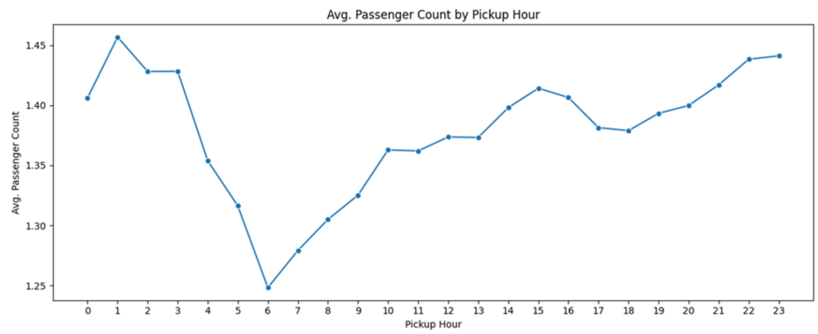


The lowest average percentage was analysed for trips more than 5 miles, passenger count 4, and pickup time around 4:00.

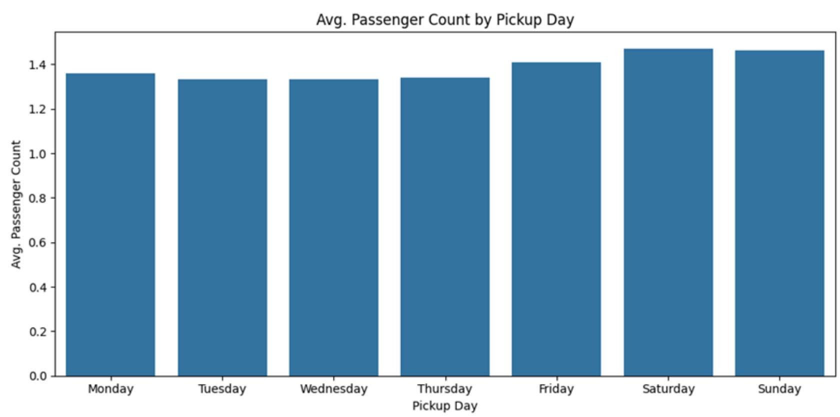
3.2.14. Analyse the trends in passenger count

Passenger count was analysed by pickup hour and pickup day. The highest average passenger count was found around 1:00 and on Saturday.

Average passenger count by pickup hour



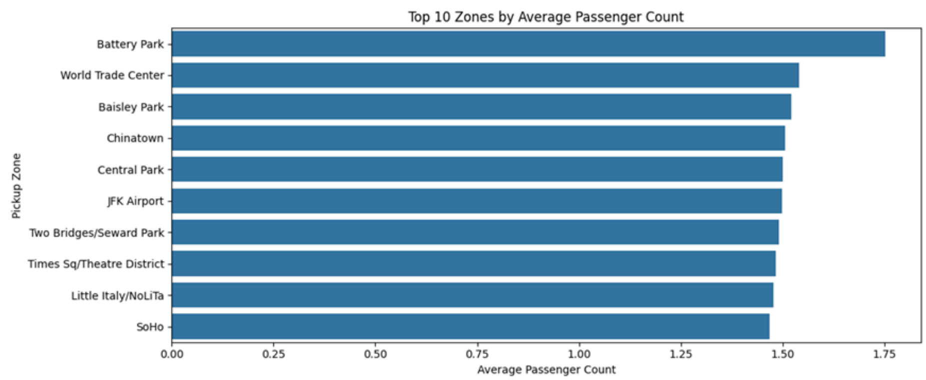
Average passenger count by pickup by day



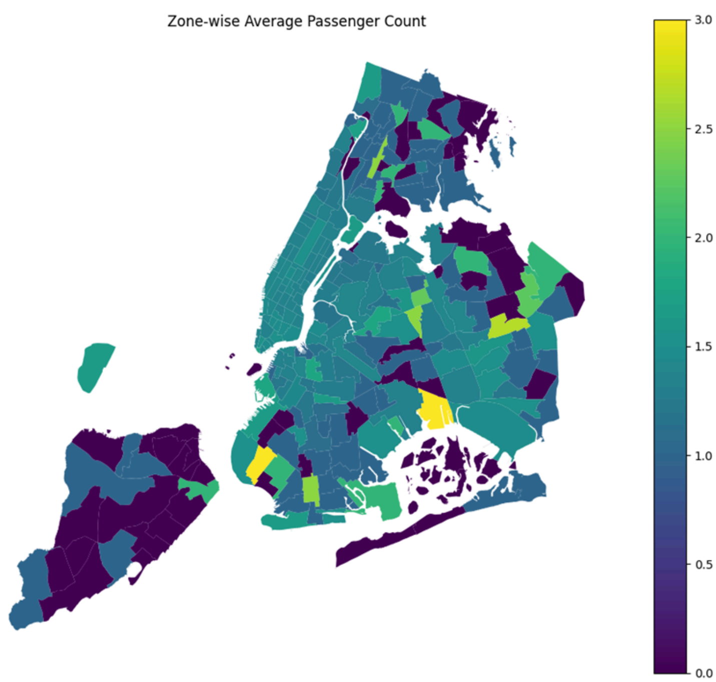
3.2.15. Analyse the variation of passenger counts across zones

Average passenger count was calculated by pickup zone and merged with the zones GeoDataFrame. Amongst the zones, 'Battery Park' had the highest passenger count, while 'Manhattanville' had the lowest.

Top zones by average passenger count



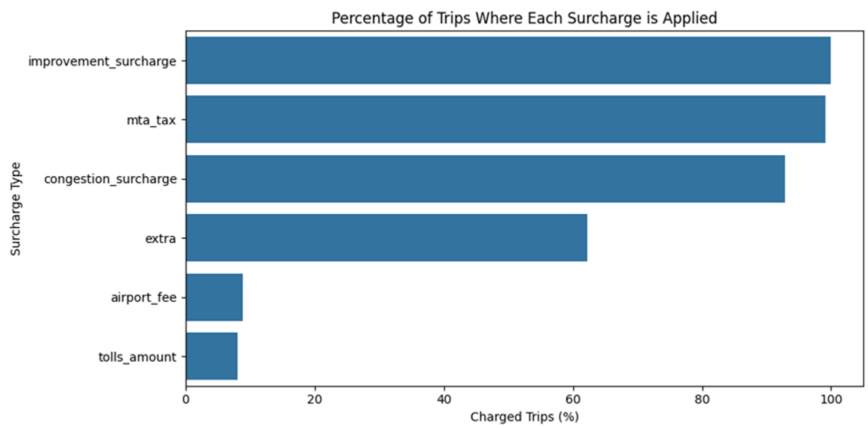
Zone-wise average passenger count map



3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

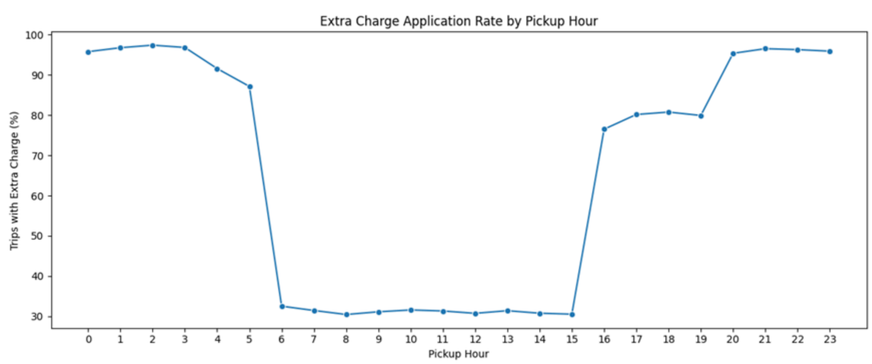
Surcharge prevalence was checked across all surcharge columns. Improvement_surchge was applied most frequently, appearing in 99.97 % of trips.

Percentage of trips where each surcharge is applied

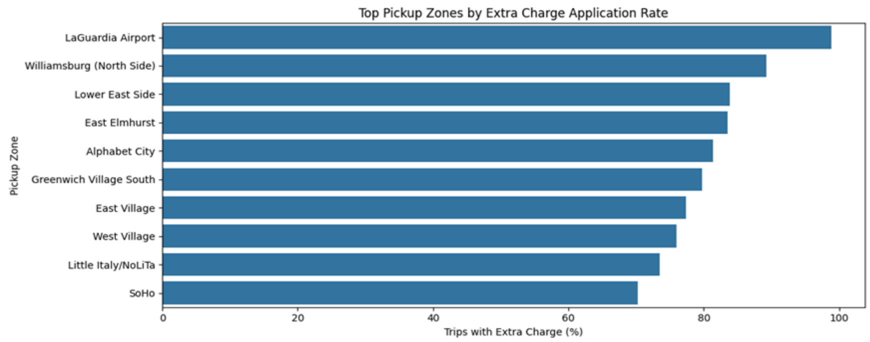


For zone/time pattern, the extra charges were most frequently around 2:00, and ‘LaGuardia Airport’ had the highest pickup and dropoff extra-charge rate.

Extra charge application rate by pickup hour



Top pickup zones by extra rate



Conclusion: The detailed EDA analysis showed that demand is highest around night 1800, selected zones, have much higher pickup/dropoff activity, and night demand is strongly visible around East villege. Pricing and fare patterns also change by passenger count and vendor, so these patterns can support better dispatching, taxi positioning and pricing decisions.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

Evening time around 18:00, shows high pickup activity, so more taxis should be available before and during this period. High pickup zones such as JFK Airport, Midtown areas and other busy pickup locations should also get more dispatch priority.

Slow routes should be monitored because they reduce the number of trips a driver can complete in a day. If some routes are repeatedly slow during specific hours, drivers can be guided through alternate routers or nearby high demand zones.

Weekday and weekend demand should be planned separately. On weekdays, more taxis should be available around high-demand pickup zones, while weekend and late night hours should focus more on night-traffic zones.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

Taxi positioning should not be random. Taxis should be placed near zones that showed high pickup demand, especially around peak hours and busy months,

Airport zones such as JFK Airport and LaGuardia Airport should have enough taxi supply because they appeared as important pickup zones. East Village should get more attention during 11 PM to 5 AM because it had the highest night pickup and dropoff traffic.

Dropoff-heavy zones can also be used for repositioning decisions. If many trips end in a zone but pickups are lower there, drivers should be guided toward nearby pickup heavy zones instead of waiting too long.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

Fare strategy should not be changed in one common way for all trips. Fare per mile changes by distance, pickup hour, day of week and vendor, so pricing should be adjusted carefully using these patterns.

Short distance trips should be analysed separately from medium and long distance trips because fare per mile is different across distance tiers. Vendor wise fare per mile should be compared before changing prices.

Daytime trips contribute most of the revenue, so prices should not be increased too aggressively during already busy periods. For nighttime, selected driver incentives or local pricing adjustments can be used in high demand night zones.